

NGILA

Community-Powered Street Economy Platform

Project Documentation

Challenge: Street Economy

Tagline: Discover your local economy.

Google Maps for the street economy — built by the community.

1. Project Information

Project Name: Ngila

Challenge: Street Economy

Project Category: Digital Platform / Mobile Application

Primary Users: Customers, street vendors, informal businesses and local communities

Project Tagline: Discover your local economy.

Alternative Pitch: Google Maps for the street economy — built by the community.

2. Executive Summary

Ngila is a community-powered, location-based mobile platform designed to make street vendors and informal businesses easier to discover, access and support.

Many street vendors provide valuable products and services within communities but have little or no digital presence. Customers often discover these businesses through word-of-mouth or by physically encountering them. This makes it difficult for new customers to know what is available around them, where vendors are located, what they sell, whether they are currently open, and whether other customers have had positive experiences.

Ngila addresses this problem by creating a digital discovery layer for the informal and street economy. Users can open the application, share their location and discover nearby vendors. They can search for products or services, filter results by distance and category, view vendor profiles, see products, ratings and reviews, and receive directions.

A key feature is community-driven vendor discovery. Customers can add vendors that are not yet registered on Ngila. The vendor can subsequently claim and verify the profile and take control of their business information.

3. Background: What Inspired the Project?

The informal and street economy plays an important role in communities by providing affordable goods, food, services and employment opportunities.

However, many street businesses remain largely invisible in the digital world. A customer may walk through a neighbourhood every day without knowing that a food vendor, fresh-produce seller, clothing trader, barber, repair service or other local business operates nearby.

Many of these businesses depend heavily on physical visibility and word-of-mouth. At the same time, customers increasingly expect to be able to search for businesses digitally.

Ngila was inspired by the opportunity to bridge this gap. Instead of expecting every informal trader to become digitally sophisticated, Ngila allows the community to help digitise the street economy.

A customer can discover a vendor and add them to Ngila. The vendor can later claim the business profile. This creates a platform where customers discover businesses, communities add businesses, vendors claim businesses, vendors become digitally visible, and more customers discover them.

4. Problem Statement

Street vendors and informal businesses face several digital challenges.

Limited Digital Visibility: Many informal businesses do not have websites, business profiles, online stores, digital advertising, accurate map listings or online reviews.

Difficulty Discovering Local Vendors: Customers often rely on word-of-mouth, walking around, asking other people, social media groups or physical signs.

Location Uncertainty: Informal businesses may not have conventional addresses. A vendor may be known as 'the person selling kota next to the taxi rank.'

Trust: Customers may be uncertain about product quality, prices, service quality and reliability. Ratings and reviews can help.

Vendor Onboarding: Requiring every vendor to independently download an application and create a profile can slow adoption. Ngila solves this through community-assisted onboarding.

5. Proposed Solution

Ngila creates a location-aware digital directory and discovery platform specifically for street vendors and informal businesses.

Customers can discover nearby vendors, search for products and services, filter vendors, view profiles, view products, view ratings and reviews, get directions, add vendors, rate vendors and save favourites.

Vendors can register, claim existing listings, verify their businesses, manage profiles, add products and prices, upload photos, update operating hours and locations, and respond to reviews.

6. Core Concept

The central idea behind Ngila is: A digital marketplace does not need to start with the businesses. It can start with the community.

A customer who discovers an excellent food vendor can add that vendor to Ngila. The vendor can then claim the profile. This dramatically lowers the barrier to digital inclusion.

7. Target Users

Customers: People looking for food, clothing, fresh produce, electronics, phone accessories, haircuts, car washes, repairs, household goods and local services.

Street Vendors: Food stalls, market stalls, mobile businesses, informal shops, clothing stalls, produce sellers, service businesses and other informal traders.

Local Communities: Communities can collectively create and maintain a digital representation of their local economy.

Future Users: Local organisations, municipalities, NGOs, community development organisations, researchers and economic development programmes.

8. Customer Experience

1. Open Ngila and share location.
2. Discover nearby vendors.
3. Search for a product or service.
4. Filter by distance, category, rating, price or opening status.
5. View a vendor profile.

6. Get directions.
7. Visit the vendor.
8. Rate and review the experience.

9. Community Vendor Discovery

If a customer encounters a vendor who is not listed, they can select 'Add Vendor' and provide the vendor name, category, products, approximate location, hours, contact details if available, photos and a description.

The listing can initially display a 'Community Added' status. The vendor can later select 'Claim this Business' and take ownership of the profile.

10. Vendor Claiming and Verification

Ngila should distinguish between community-created listings and verified business profiles.

Community Added — created by a customer.

Vendor Claimed — the vendor has claimed the listing.

Verified Vendor — the business has completed verification.

11. Vendor Profiles

A vendor profile can contain the business name, rating, distance, current opening status, products, prices, description, operating hours, photos, location and actions such as directions, call, share, favourite, rate and report.

12. Product Discovery

Vendors should eventually be able to list individual products and prices. For example: Kota R35, Chips R20, Chicken R30 and Coke R15.

13. Search and Filtering

Users can search by product, service, category, vendor name or nearby location.

Filters can include distance, category, rating, price, open/closed status and verified status.

14. Location-Based Discovery

Location is fundamental to Ngila. The application uses GPS coordinates to find nearby vendors. Users can select a radius such as 500 m, 1 km, 2 km, 5 km or 10 km.

15. Maps and Directions

Users can select 'Get Directions' to navigate to a vendor. This is especially useful because informal businesses may not have conventional street addresses. Ngila can work with GPS coordinates and descriptions such as 'next to the taxi rank.'

16. Ratings and Reviews

Customers can rate vendors from 1–5 stars and leave written reviews. Optional tags can include Good Service, Affordable, Friendly, Good Quality, Good Food, Fast Service and Convenient Location.

17. Vendor Reputation

Ngila can develop reputation indicators such as Highly Rated, Popular, Trusted, Community Favourite and Verified.

18. Community Moderation

Users should be able to report fake businesses, incorrect locations, duplicate listings, incorrect information, offensive content, fraudulent reviews and closed businesses. Administrators can investigate reports.

19. Gamification

Users can earn points for adding vendors, writing reviews, uploading photos, correcting information, confirming locations and reporting inaccurate information.

Potential levels include Explorer, Contributor, Community Builder and Ngila Ambassador.

20. Smart Recommendations

A future version can provide personalised recommendations based on user preferences, previous searches, favourite vendors, distance, time of day, ratings and popularity.

21. Future AI Features

Natural-language search could allow users to say, 'I want something cheap to eat within walking distance.'

AI could recommend vendors, analyse reviews for common themes and help vendors create business descriptions, product descriptions and promotional content.

22. Vendor Dashboard

A vendor dashboard can show profile views, directions requested, phone calls, favourites, ratings and most-viewed products. This gives informal businesses useful business insights.

23. Future Marketplace

Ngila could eventually evolve from discovery into transactions: Discover → View Product → Order → Pay → Collect/Deliver. This should be a later-stage feature rather than part of the first MVP.

24. Future Digital Payments

Later versions could integrate digital payments, mobile payments, payment links and QR payments.

25. Business Model

Potential revenue streams include sponsored listings, featured vendors, premium vendor profiles, advertising, advanced analytics and partnerships with financial institutions, mobile networks, municipalities and development programmes.

26. Social Impact

Ngila can increase visibility, help businesses attract customers, support local economies, digitise informal businesses, create trust through reviews and generate useful aggregated insights into local economic activity.

27. Network Effect

More customers lead to more vendors being added. More vendors lead to more information and better search results. Better search results make the platform more useful, attracting more customers.

More vendors also create more products and reviews, which increase trust and further attract customers.

28. Technical Architecture

Mobile Application: React Native + TypeScript for the interface, GPS/location, maps, camera, vendor discovery, reviews, authentication and notifications.

Backend: Django + Django REST Framework for authentication, user management, vendor management, products, reviews, ratings, search, filtering and verification.

Database: PostgreSQL, with PostGIS recommended for geographic queries such as finding vendors within a specified distance.

Maps: Google Maps Platform or Mapbox.

Authentication: JWT-based authentication.

Storage: Cloud storage for vendor and product images.

Notifications: Push notifications for updates, reviews, verification and future promotions.

29. Core Database Entities

Suggested entities: User, Vendor, VendorClaim, VendorLocation, Category, Product, Review, Rating, Favourite, VendorPhoto, OperatingHours, Report and Notification.

30. Suggested API Structure

Authentication: `/api/auth/register/` and `/api/auth/login/`

Vendors: `/api/vendors/`, `/api/vendors/{id}/`, `/api/vendors/nearby/`, `/api/vendors/search/`, `/api/vendors/{id}/products/`, `/api/vendors/{id}/reviews/` and `/api/vendors/{id}/claim/`

Products and categories: `/api/products/` and `/api/categories/`

Reviews and ratings: `/api/reviews/` and `/api/ratings/`

Favourites: `/api/favourites/`

Reports: `/api/reports/`

31. MVP — Minimum Viable Product

Customer MVP: Registration/login, location detection, nearby vendors, search, category and distance filtering, profiles, products, ratings, reviews, directions, add vendor and favourites.

Vendor MVP: Registration, claim vendor, verification, profile management, product management, location management and operating hours.

Admin MVP: User management, vendor management, verification, categories, reports, fraudulent listing removal and review moderation.

32. MVP User Flow

Open Ngila → Share Location → Nearby Vendors → Search/Filter/Map → Vendor Profile → Products/Reviews/Location → Directions → Visit → Review.

33. Community Vendor Flow

Customer discovers vendor → Add Vendor → Enter basic information → Community listing → Other users rate/review → Vendor discovers profile → Claim business → Verification → Vendor controls profile.

34. Vendor Lifecycle

Community Added → Pending Verification → Vendor Claimed → Verified → Active.

Future states can include Temporarily Closed, Moved, Inactive, Permanently Closed and Suspended.

35. Security Considerations

Security measures should include secure authentication, password hashing, JWT expiration, API permissions, vendor ownership verification, review moderation, rate limiting, input validation, image validation, location privacy controls, abuse reporting and role-based admin permissions.

36. Privacy Considerations

Because Ngila uses location, the application should request permission before accessing location, explain why it is required, avoid unnecessarily storing exact customer locations, minimise retained location data, allow users to disable location access, protect personal information and consider applicable privacy requirements such as POPIA.

37. Scalability

Ngila should be designed to grow from one community to one city, multiple cities, South Africa and potentially other countries. Geographic data, vendor data, products, reviews and users should be structured for future scaling.

38. South African Expansion

The platform could initially focus on one geographic area and later expand to Johannesburg, Pretoria, Soweto, Midrand, Cape Town, Durban, Gqeberha, Polokwane, Mbombela and other regions.

39. Competitive Differentiation

Ngila should not simply become another general business directory. Its focus should be the informal and street economy, especially businesses that may lack formal addresses, websites, online stores, digital marketing, existing map profiles or technical knowledge.

40. What Makes Ngila Innovative?

1. Community-created businesses.
2. Vendor claiming.
3. Location-first discovery.
4. Informal-economy focus.
5. Community reputation.
6. Digital inclusion.
7. Geographic intelligence through location-aware search.

41. Example Real-World Scenario

A university student is hungry after class. They open Ngila, share their location, search for Food, select Within 1 km and Open Now, and discover Mama Mary's Kitchen 350 m away with a 4.8 rating. They view the menu, choose Get Directions, visit the vendor and later leave a five-star review. Another customer subsequently discovers the same vendor through that review.

42. Long-Term Vision

The long-term vision is for Ngila to become a digital infrastructure layer for the informal economy, providing discovery, navigation, reputation, commerce, analytics, financial inclusion and business-growth opportunities.

43. Development Roadmap

Phase 1 — Foundation: Project setup, Django backend, PostgreSQL/PostGIS, React Native application, authentication, user accounts and basic APIs.

Phase 2 — Vendor System: Vendor model, categories, registration, profiles, products and locations.

Phase 3 — Discovery: GPS, nearby vendors, radius search, search, filtering and map integration.

Phase 4 — Community: Add vendor, vendor claiming, verification, ratings, reviews, photos and reporting.

Phase 5 — Engagement: Favourites, notifications, gamification and contributor profiles.

Phase 6 — Intelligence: Recommendations, natural-language search, review analysis and vendor analytics.

Phase 7 — Commerce: Product ordering, digital payments, promotions and delivery/collection.

44. Recommended MVP Scope

The first working prototype should prioritise user authentication, GPS location, vendor database, nearby vendor search, maps, category filtering, vendor profiles, products, ratings, reviews, add vendor, vendor claiming and directions.

45. Project Vision Statement

Ngila aims to make every local street business discoverable, accessible and connected to the community around it.

46. Mission Statement

To empower communities to digitally discover, support and grow the informal businesses that form an essential part of their local economy.

47. Final Project Pitch

Ngila is a community-powered mobile platform for discovering street vendors and informal businesses. By combining location-based discovery, community-generated vendor listings, vendor verification, product information, ratings, reviews and navigation, Ngila connects customers with businesses that are often invisible in the digital economy.

Unlike traditional business directories that depend on businesses creating their own digital presence, Ngila allows customers to add vendors they discover. Vendors can then claim and manage their profiles. This creates a continuously expanding digital map of the street economy, built collectively by the communities that participate in it.

Ngila's vision is to make local informal businesses as discoverable and accessible as formal businesses, while creating new opportunities for digital inclusion, customer engagement and economic growth.

48. One-Line Description

Ngila connects people to the street economy around them.

49. Short Elevator Pitch

Ngila is a location-based platform that helps people discover street vendors and informal businesses nearby. Customers can search, filter, navigate to, rate and review vendors, while also adding businesses that aren't yet on the platform. Vendors can then claim and manage their profiles. In this way, Ngila creates a community-powered digital map of the informal economy.

50. Core Philosophy

Find it. Support it. Rate it. Grow it.

The platform is not just about finding vendors. It is about making the invisible visible. It gives customers a way to discover the businesses around them and gives informal businesses a path into the digital economy without requiring them to already be digitally established.